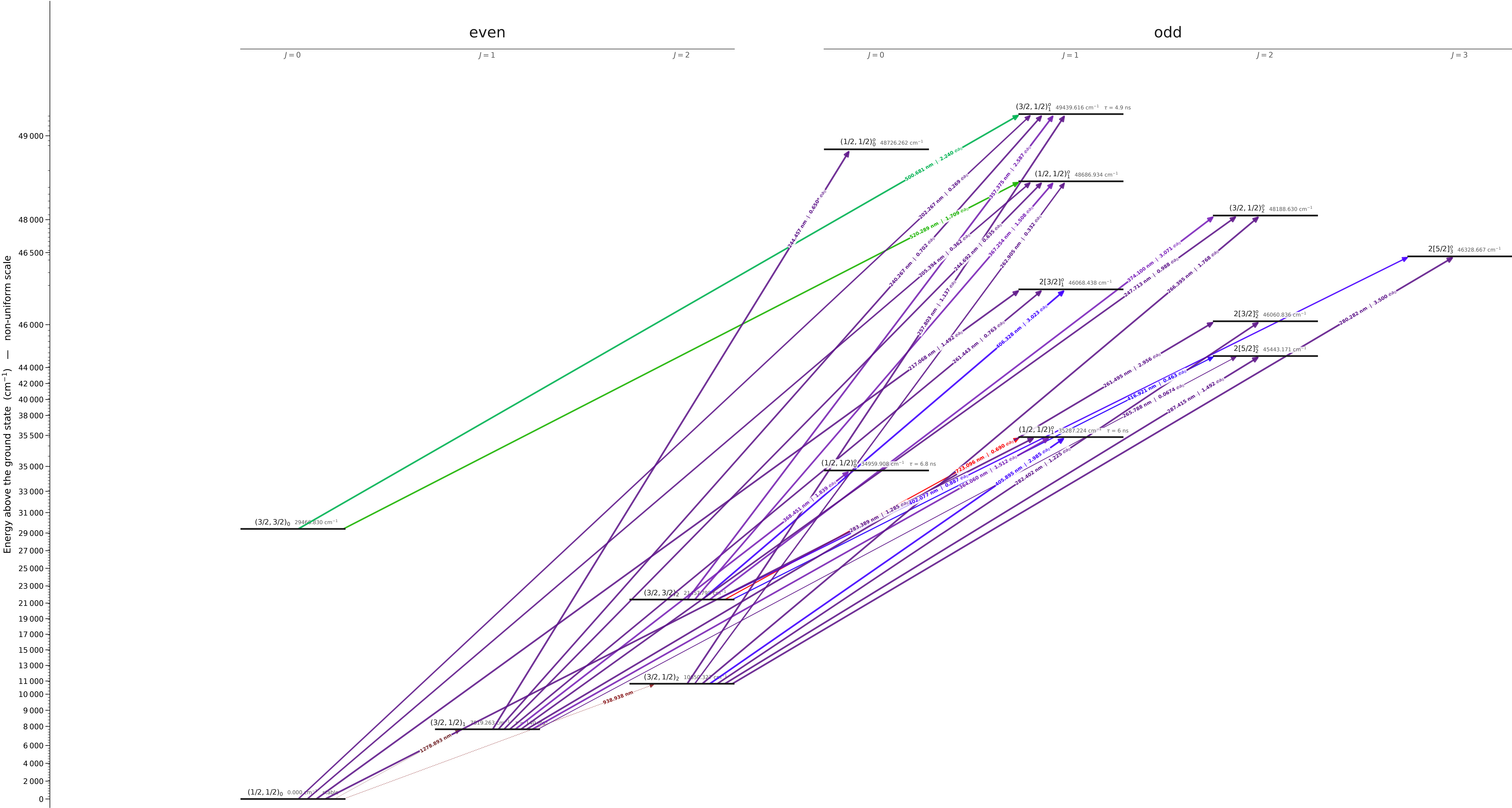
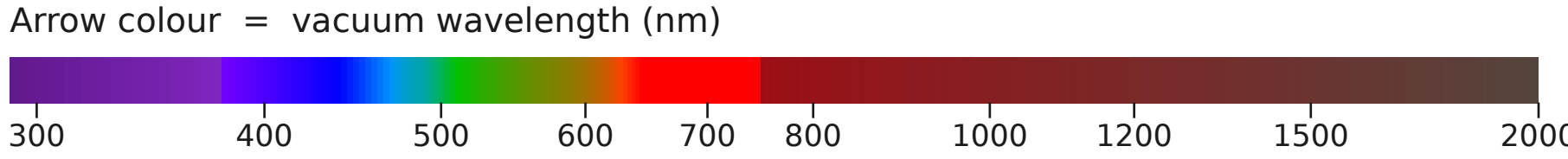


Lead

Pb (Pb I). Strongest classified transitions below 2 μm (NIST): 15 levels, 31 lines.

Pb + ionisation limit 59819.56 cm⁻¹ = 7.41668 eV (167.169 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[J||e\mathbf{r}||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	[J e $\mathbf{r}$ J'] (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
(1/2, 1/2) ₀ – (1/2, 1/2) ₁ ^o	283.3887	283.3053	1057.8844	1.285	4.900e+07	26.53	29.4	ground-state resonance line
(3/2, 1/2) ₁ – (1/2, 1/2) ₁ ^o	364.0605	363.9568	823.4688	1.512	3.200e+07	26.53	19.2	isotope shifts and hyperfine structure measured by Dopple...
(3/2, 1/2) ₁ – (1/2, 1/2) ₀ ^o	368.4511	368.3462	813.6561	1.839	1.370e+08	23.41	100	proposed laser-cooling / magneto-optical-trap cycling line
(3/2, 1/2) ₂ – (1/2, 1/2) ₁ ^o	405.8953	405.7807	738.5956	2.985	9.000e+07	26.53	54	the Pb I line used for lead abundances in stellar spectra
(1/2, 1/2) ₀ – (3/2, 1/2) ₂	938.9383	938.6807	319.2888	-	-	-	-	electric-quadrupole line measured by Faraday-rotation spe...
(1/2, 1/2) ₀ – (3/2, 1/2) ₁	1278.8930	1278.5432	234.4156	-	7.190e+00	1.137e-06	100	magnetic-dipole line used for parity-nonconservation opti...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁰⁷Pb hyperfine structure (I = 1/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
(3/2, 1/2) ₁	-2389.4(7)	-	F=1/2: +2389.40 F=3/2: -1194.70
(3/2, 1/2) ₂	2600.8(9)	-	F=3/2: -3901.20 F=5/2: +2600.80
(3/2, 3/2) ₂	609.820(8)	-	F=3/2: -914.73 F=5/2: +609.82
(1/2, 1/2) ₁ ^o	8802.0(16)	-	F=1/2: -8802.00 F=3/2: +4401.00
2[5/2] ₂ ^o	3094(9)	-	F=3/2: -4641.00 F=5/2: +3094.00
2[3/2] ₂ ^o	-1456.000	-	F=3/2: +2184.00 F=5/2: -1456.00
2[3/2] ₁ ^o	-2884.000	-	F=1/2: +2884.00 F=3/2: -1442.00
2[5/2] ₁ ^o	2072(8)	-	F=5/2: -4144.00 F=7/2: +3108.00
(3/2, 1/2) ₂ ^o	1734.000	-	F=3/2: -2601.00 F=5/2: +1734.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
(1/2, 1/2) ₀ – (1/2, 1/2) ₁ ^o	283.389	204-208	-4212.0(25)
(1/2, 1/2) ₀ – (1/2, 1/2) ₁ ^o	283.389	206-208	-2226.6(25)
(1/2, 1/2) ₀ – (1/2, 1/2) ₁ ^o	283.389	207-208	-1390.0(25)
(3/2, 1/2) ₂ – (1/2, 1/2) ₁ ^o	405.895	206-208	-2428.00
(3/2, 1/2) ₂ – (1/2, 1/2) ₁ ^o	405.895	204-208	-4587.00
(3/2, 1/2) ₂ – (1/2, 1/2) ₁ ^o	405.895	207-208	-1569.00
(1/2, 1/2) ₀ – (3/2, 1/2) ₁	1278.893	206-208	220.4(15)
(1/2, 1/2) ₀ – (3/2, 1/2) ₁	1278.893	207-208	138.0(17)

Shift = ν(first isotope) – ν(second isotope).